Real-shift coefficientwise log-concavity for rising-factorial ratios

A proof of MathDB problem 324472

Principia Math In Progress

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Task 1: solve the original problem. We prove Conjecture 1 of Kalmykov–Karp [1, p. 408] under the hypotheses of their Theorem 1, with an additional relaxation when the parameter gap is at most one.

Theorem 1. *Let $c \geq a > 0$, and let f_j be a nonnegative log-concave sequence without internal zeros. Define*

$$F(\mu; x) = \sum_{j \geq 0} f_j \frac{(a+\mu)_j}{(c+\mu)_j} \frac{x^j}{j!}, \qquad (b)_j = \prod_{i=0}^{j-1} (b+i).$$

For every $\mu, \nu \geq 0$, every coefficient of

$$F(\mu; x)F(\nu; x) - F(0; x)F(\mu+\nu; x) \tag{1}$$

is nonnegative. If $0 \leq c-a \leq 1$, the same conclusion holds for every nonnegative sequence f_j, without the log-concavity restriction. These are formal-series assertions; they imply the corresponding Wright log-concavity for nonnegative x in the common convergence domain.

The proof uses elementary comparisons of densities. We recall the one-crossing argument: if two probability densities on an interval have an increasing ratio, then the second law is stochastically larger, provided the ratio is second divided by first. Indeed the difference of the normalized densities crosses zero at most once, from negative to positive, so their distribution functions have the required order. The same argument applies if the second support also extends farther to the right and has no extra portion to the left. Decreasing ratios reverse the conclusion.

Lemma 2. *For symmetric nonnegative weights satisfying*

$$w_k = w_{n-k}, \qquad w_0 \leq w_1 \leq \cdots \leq w_{\lfloor n/2 \rfloor}, \tag{2}$$

the Bernstein polynomial

$$R_w(r) = \sum_{k=0}^{n} \binom{n}{k} w_k r^k (1-r)^{n-k} \tag{3}$$

is nondecreasing on $[0, 1/2]$ and satisfies $R_w(r) = R_w(1-r)$.

Proof. Symmetry is immediate. For $n \geq 1$, differentiate (3):

$$R_w'(r) = n \sum_{k=0}^{n-1} (w_{k+1} - w_k) \binom{n-1}{k} r^k (1-r)^{n-1-k}. \tag{4}$$

The coefficient differences at reflected indices are opposite, with the lower-index one nonnegative. On $[0, 1/2]$, the lower-index Bernstein term is at least its reflection. Pairing terms proves (4) nonnegative. The cases $n = 0, 1$ are constant. □

Lemma 3. *Let $X \sim \mathrm{Beta}(u,d)$ and $Y \sim \mathrm{Beta}(v,d)$ be independent, with $u,v,d>0$, and put*

$$P_w(x,y)=\sum_{k=0}^{n}\binom{n}{k}w_k x^k y^{n-k}. \tag{5}$$

At fixed $S=u+v$, the expectation $\mathbb{E}P_w(X,Y)$ is nondecreasing as u,v are balanced, in either of the following cases:

1. *$0<d\le 1$ and w is any symmetric nonnegative vector;*
2. *$d>1$ and w satisfies (2).*

Proof. Write

$$q=\frac{\min(X,Y)}{\max(X,Y)},\qquad z=\max(X,Y),\qquad \delta=\frac{|u-v|}{2}.$$

Both q,z lie in $(0,1)$ with probability one. Adding the densities from the two possible orders of X,Y, with Jacobian z, gives the joint density

$$\frac{2}{B(u,d)B(v,d)}z^{S-1}q^{S/2-1}(1-z)^{d-1}(1-qz)^{d-1}\cosh(\delta\log q). \tag{6}$$

In particular the conditional law of z given q does not depend on δ. The marginal density of q, apart from its normalizing constant, has the form

$$A(q)q^{S/2-1}\cosh(\delta\log q),\qquad A(q)=\int_0^1 z^{S-1}(1-z)^{d-1}(1-qz)^{d-1}\,dz. \tag{7}$$

The integral is positive and finite for $0<q<1$. If $\delta_2\ge\delta_1\ge 0$, the ratio of the marginal densities is decreasing in q. To check this, put $t=-\log q>0$ and note

$$\frac{d}{dt}\log\frac{\cosh(\delta_2 t)}{\cosh(\delta_1 t)}=\delta_2\tanh(\delta_2 t)-\delta_1\tanh(\delta_1 t)\ge 0. \tag{8}$$

Thus q is stochastically larger as δ decreases. Since P_w is symmetric, it remains to show that the δ-independent conditional mean

$$m(q)=\mathbb{E}[P_w(X,Y)\mid q] \tag{9}$$

is nondecreasing in q.

Suppose first that $0<d\le 1$. For $0<q_1<q_2<1$, the ratio of the conditional z-densities, apart from a positive constant, is

$$\left(\frac{1-q_2 z}{1-q_1 z}\right)^{d-1}. \tag{10}$$

The inner ratio decreases in z, so (10) increases in z. Consequently the conditional moment $\mathbb{E}[z^n\mid q]$ is nondecreasing in q. Also,

$$P_w(z,qz)=z^n Q_w(q),\qquad Q_w(q)=\sum_{k=0}^{n}\binom{n}{k}w_k q^{n-k}. \tag{11}$$

The polynomial Q_w has nonnegative coefficients, hence is nonnegative and nondecreasing on $(0,1)$. Formula (11) proves the required monotonicity of $m(q)$.

Now suppose $d>1$, and set $T=X+Y=(1+q)z$. By (6), its conditional density given q, apart from a positive normalizing constant, is

$$t^{S-1}\bigl(1-t+h(q)t^2\bigr)^{d-1},\quad 0<t<1+q,\qquad h(q)=\frac{q}{(1+q)^2}. \tag{12}$$

The quadratic is the product $(1 - t/(1+q))(1 - qt/(1+q))$, so it is positive on that support. Moreover h is increasing for $0 < q < 1$. If $q_2 > q_1$, write $h_i = h(q_i)$. On the common support the ratio of the two quadratics is

$$\frac{1 - t + h_2 t^2}{1 - t + h_1 t^2} = 1 + (h_2 - h_1)\frac{t^2}{1 - t + h_1 t^2}. \tag{13}$$

It is increasing with t, since

$$\frac{d}{dt}\frac{t^2}{1 - t + h_1 t^2} = \frac{t(2-t)}{(1 - t + h_1 t^2)^2} > 0 \quad (0 < t < 1 + q_1 < 2). \tag{14}$$

Raising (13) to $d - 1 > 0$ preserves this monotonicity, and the upper support endpoint $1 + q$ increases. The one-crossing argument therefore shows that T, conditionally on q, is stochastically increasing with q. In particular $\mathbb{E}[T^n \mid q]$ increases.

Homogeneity, symmetry, and Lemma 2 give

$$P_w(X, Y) = T^n R_w\left(\frac{q}{1+q}\right). \tag{15}$$

The second factor is nonnegative and nondecreasing in q, since $q/(1+q) \in (0, 1/2)$. Multiplying it by $\mathbb{E}[T^n \mid q]$ proves that (9) is nondecreasing. In both cases, the stochastic comparison following (8) now proves the lemma.

All conditional-density arguments take place on $0 < q < 1$. Possible limiting singularities at $q = 1$ cause no exception: that endpoint has probability zero, and the polynomial P_w is bounded on $[0, 1]^2$. □

Proof of Theorem 1. If $d = c - a = 0$, every rising-factorial ratio is one and (1) vanishes. Assume $d > 0$. For $B_{u,d} \sim \mathrm{Beta}(u, d)$, the beta integral gives

$$\mathbb{E}B_{u,d}^k = \frac{B(u+k, d)}{B(u, d)} = \frac{(u)_k}{(u+d)_k}. \tag{16}$$

For each degree n, take $w_k = f_k f_{n-k}$ in (5). The Cauchy product and (16) identify $n!$ times the coefficient of x^n in (1) with

$$\mathbb{E}P_w(B_{a+\mu,d}, B_{a+\nu,d}) - \mathbb{E}P_w(B_{a,d}, B_{a+\mu+\nu,d}), \tag{17}$$

where the two beta variables in each expectation are independent. The first parameter pair is at least as balanced as the second, because their sums agree and $|\mu - \nu| \le \mu + \nu$.

If f is log-concave without internal zeros, its product weights satisfy (2). On the positive support this follows from

$$\frac{w_{k+1}}{w_k} = \frac{f_{k+1}/f_k}{f_{n-k}/f_{n-k-1}} \ge 1 \quad (k < n/2). \tag{18}$$

Its positive support is a centered interval, so initial or terminal zeros cause no exception. If all weights vanish, the coefficient is zero. Lemma 3 proves (17) nonnegative. Its first case also proves the stated extension for arbitrary nonnegative f when $d \le 1$. This covers degree zero and vanishing shifts as well. Replacing (a, c) by $(a + t, c + t)$, $t \ge 0$, supplies every common base shift required in the Wright definition. □

Verification. The companion program passes 441 exact rational coefficient cases, 3 exact density-derivative identities, 225 conditional-moment comparisons, and 3 independent transformed beta integrals. The analytic proof, not these finite checks, establishes all degrees and all parameters. Source and current claim checks are recorded in the literature audit.

Task 2. Determine when arbitrary nonnegative input can produce a negative coefficient for $c - a > 1$, and find the smallest failing degree as a function of the parameter gap and the base parameter.

Task 3. Determine all equality cases coefficient by coefficient and derive sharp quantitative lower bounds for (1) for arbitrary positive real shifts.

Task 4. Extend the conditional-density comparison to products of three or more beta variables and identify multivariate coefficient conditions that give monotonicity under majorization of the shape parameters.

Task 5. Prove or refute analogous real-shift coefficient inequalities for series containing several rising-factorial ratios, using joint ratio conditioning to handle the additional parameter differences.

References

[1] S. I. Kalmykov and D. B. Karp, Log-convexity and log-concavity for series in gamma ratios and applications, *J. Math. Anal. Appl.* **406** (2013), 400–418. doi:10.1016/j.jmaa.2013.04.061; arXiv:1211.2882.